

A lot of companies have also started manufacturing their own desktop 3D printers. Talking about the launch of Stratasys' new printer in July, 2014, Rodagi says, "The Connex3 colour multi-material 3D printer series features a unique triple-jetting technology that combines droplets of three base materials to produce parts with virtually unlimited combinations of rigid, flexible and transparent colour materials, as well as colour digital materials, all in a single print run."

Progress in industrial 3D printing technology

Significant development has happened in the last couple of years in the industrial 3D printing sector. Industrial 3D printers that use materials like titanium and other highly-refined materials can print new and replacement parts for aerospace, automobile and medical device industries. Mishra says, "A more refined workflow system is helping many companies to use 3D-printing processes to offer distributed-parts delivery systems, enabling them to cut cost in terms of inventory and product upgradation." On personal printers front, he adds, "Many companies have now started offering products on reliable FDM platforms with low-cost printers (starting US\$499) and making these affordable for professional as well as home use."

Coming years will see 3D-printing materials get more suitable and compatible with injection-moulded parts, feels Mishra. Consequently, the cost of these 3D-printed parts would fall down. He says, "It will also see a huge technological upgradation in terms of productivity and printed sizes to allow companies to use 3D-printed objects in their final production process."

Technology gets better by the day. Big names, like Autodesk and HP, have decided to enter the industrial 3D-printing market. Canon is expected to follow shortly. "Though

Developments in desktop 3D printing: In a nutshell

Increase in the number of commercially-available desktop 3D printers. Earlier, names like Makerbot and Ultimaker dominated the market. However, entry of Chinese companies in this arena has brought costs down drastically.

Developments in software. Another notable development was the movement by 3D-printer makers towards connecting 3D printers with easy-to-use apps. Earlier desktop printers used open source software to generate G-codes and print objects. With rapid development in the field of hardware, there is equal development in the field of software. App based 3D printing makes it easy and fun for newbies to print objects on their desktop 3D printers. Windows 8.1 also provides an in-built 3D-printing app.

Increase in the choice of materials. Initial FDM based desktop 3D printers used poly-lactic acid (PLA) and acrylonitrile-butadiene-styrene (ABS) to print objects. Now, a larger choice is available to users—nylon, flexible filament, wood-texture filament and water-soluble filament being some examples. Some established companies have also launched pure transparent filaments and multi-colour filaments. Although for industrial printers, the choice of materials is unlimited and ranges from metals like gold, copper, silver to ceramics, carbon fibre and many more.

Increase in web resources. A large number of web based resources for sharing 3D-printing models have mushroomed. Thingiverse.com is one of the most popular ones. Thousands of functional models that touch our lives daily are downloaded and uploaded every day on such websites, thus making the common man a part of this disruptive technological revolution.

—Inputs from Vishesh Shishodia, co-founder, 3DPrintronics

these are high-end and not affordable for hobbyists or small/medium scale companies, there are a lot of start-ups focussed on delivering qualitative machines at affordable prices," says Surendranath Reddy, founder and CEO, Redd Robotics and 3D-ing. He adds, "Particularly in India, with the Make In India movement catching up, we are expecting exports of 3D printers to happen in a big way."

Interesting activities triggered by 3D printing

In medical field. 3D-printing technology is a very rapidly developing field. Every day we hear of new things that are being done with the help of 3D printing. Shishodia says, "Printing computerised tomography (CT) scans of patients provide doctors with exact moulds required for producing transplant parts. A patient was recently transplanted with a 3D-printed titanium facial implant. There are research labs working on printing complete organs using 3D printers by using amino acids and other life-forming chemicals, such as ink." He adds, "A home project started by a father by printing a prosthetic hand for

his son has taken the shape of a massive revolution of 3D-printed prosthetics."

In space. Talking about the world of space exploration, imagine a day when an astronaut is on Mars and something breaks. What does he or she do when in urgent need of a replacement? In space, where there is no comfort of overnight shipping, to be able to make parts on demand will forever change that situation for astronauts. With 3D printers, an astronaut has the ability to make whatever is needed, right there on the spot.

For instance, recently NASA printed tools on International Space Station (ISS), shares Shishodia. He says, "You transmit something from somewhere and it appears somewhere else within a few minutes." The 3D printer printed a tool with a design file transmitted from the ground to the printer at ISS. The tool was a ratchet wrench.

In consumer sector. 3D printing is helping convert ideas to physical products faster than expected. The evolving eco-system is allowing product designers and marketers to market their products faster and enabling the product delivery process, informs